ILLINOIS INSTITUTE OF TECHNOLOGY IPRO 497-615

Smart Occupancy & Emergency Response System

Soteria

Objective

Develop a PoE-powered system that uses sensors and smart automation to provide real-time, energy-efficient emergency response with guided LED evacuation.

Problem Statement

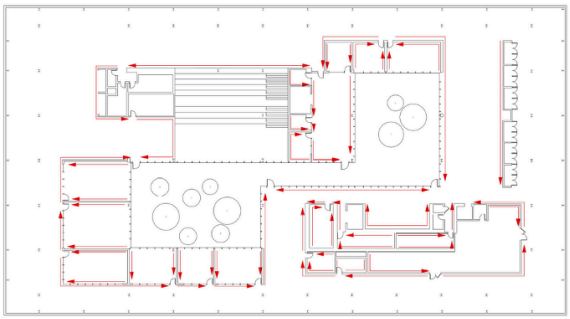
Current systems lack real-time occupancy detection and visual guidance, leading to slower evacuations and reduced responder awareness.

Project Solution

Create a scalable emergency response framework for schools, hospitals, and commercial buildings.

Technologies Used

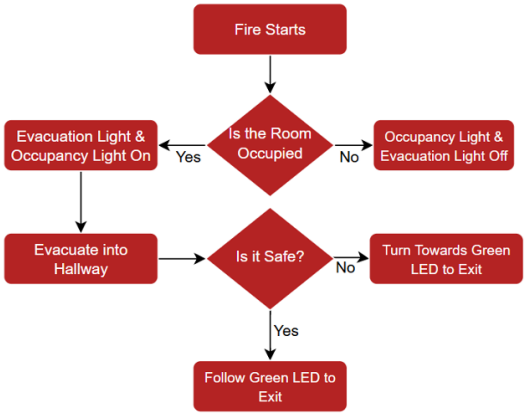
- ESP32 Microcontroller
- Breadboard
- Jumper Wires
- HLK-LD2410C Human Presence Radar Sensor
- DHT22 Temperature & Humidity Sensor
- MQ135 Air Quality Sensor
- WS2812B LED Strip



Key Benefits

- Guides people to safety in real time using LED lights that respond to room occupancy
- Saves energy with smart lighting that only turns on when needed, powered by PoE
- Easily expands to fit schools, hospitals, offices, and other buildings

How It Works



Team Responsibilities

- **Hardware Team:** Component selection, physical build, wiring design
- **Software Team:** Python programming, data handling, server integration, UI creation
- **Marketing Team:** Poster & business proposal